

Integrating Neurosymbolic AI for Pneumonia Diagnosis Using Chest X-Rays: A Synergistic Approach

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Abstract—Pneumonia remains a major global health concern, and early diagnosis from chest X-rays is essential for timely treatment. While Convolutional Neural Networks (CNNs) have demonstrated strong performance in medical image analysis, their limited interpretability often reduces clinical trust and adoption. In this work, we propose a neurosymbolic AI framework that combines CNN based image prediction with symbolic reasoning derived from structured clinical symptoms and image based features, integrating data-driven learning with interpretable rule based reasoning to improve both diagnostic accuracy and transparency. The proposed framework was evaluated on 4,277 chest X-ray images from the PadChest dataset, containing balanced pneumonia and non pneumonia cases. Six CNN architectures were trained to classify chest X-rays, and Local Interpretable Model agnostic Explanations (LIME) and Gradient weighted Class Activation Mapping (Grad-CAM) were applied to identify salient regions and gradient based cues influencing model predictions. LIME provides localized, feature based explanations by approximating the model's decision boundary around individual predictions, while Grad-CAM generates class discriminative heatmaps that visualize diagnostically relevant regions in chest X-ray images. Symbolic rules reflecting clinical symptom patterns were generated using decision tree based algorithms and integrated with CNN outputs through a weighted ensemble strategy. The resulting neurosymbolic model achieved 78% accuracy, representing an 8% improvement over the EfficientNetB0 baseline, while reducing false negatives by 19%. Interpretability also improved substantially: attention accuracy reached 0.83 IoU, rule traceability remained at 100%, and clinical validation by three medical doctors using a structured 10 point Likert scale yielded an explanation plausibility score of 8.9/10 with 89% clinical agreement. These findings demonstrate that neurosymbolic AI can effectively enhance both predictive performance and interpretability in medical imaging, providing a reliable decision support tool for AI-assisted pneumonia diagnosis.

Index Terms—Neurosymbolic AI, Pneumonia diagnosis, Chest X-ray analysis, Convolutional Neural Networks, Rule based systems, Explainable AI, Hybrid AI models

I. INTRODUCTION

Pneumonia remains a leading cause of morbidity and mortality worldwide, particularly among vulnerable populations such as young children and the elderly [12]. Early and accurate diagnosis is critical for preventing complications and enabling timely treatment. Chest X-rays are the standard diagnostic

tool for identifying lung abnormalities, but their interpretation depends heavily on radiologists' expertise, which can vary and is often limited by global shortages of specialists [5]. This has motivated the development of automated, reliable AI tools to support faster and more consistent pneumonia detection.

Convolutional Neural Networks (CNNs) have demonstrated strong performance in medical image analysis, including pneumonia detection from chest X-rays [1]. Transfer learning, which adapts models pre trained on large datasets to smaller medical datasets, has further improved performance when labeled images are limited [2]–[4]. For instance, CheXNet has reported high predictive performance ($AUC \sim 0.99$) [5]. However, CNNs often function as black boxes, providing little insight into how predictions are made, which can limit clinical trust and adoption [7]. Moreover, these models lack explicit reasoning capabilities, which are important for clinicians who need interpretable explanations of diagnostic decisions.

Neurosymbolic AI has recently emerged as a hybrid approach that combines the pattern recognition strengths of neural networks with the logical reasoning capabilities of symbolic AI [7]. Prior work in medical imaging has explored integrating CNNs with radiomic features or image derived rules [2], [3], but such approaches still often include image derived information in the symbolic component, which can complicate interpretability. In contrast, our approach separates these responsibilities: CNNs are used exclusively for visual pattern recognition, capturing features such as lung opacities and consolidations, while symbolic classifiers encode domain knowledge and decision rules from structured clinical annotations, including patient symptom columns present in the PadChest dataset. This separation enhances both transparency and clinical interpretability [11].

In this study, we propose a neurosymbolic ensemble framework that combines six state of the art CNN architectures with rule based symbolic classifiers in a weighted ensemble. The system is evaluated on the PadChest dataset, a large, expertly annotated repository of chest X-rays [14]. Performance is assessed using standard metrics, such as accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC-ROC) [12], reported as mean $\pm$ standard

deviation over multiple runs with different random seeds. For interpretability, Local Interpretable Model agnostic Explanations (LIME) [13] and Gradient weighted Class Activation Mapping (Grad-CAM) [8], [9] are applied to explain the decision rationale for both standalone CNN models and the hybrid ensemble, highlighting features and rules influencing predictions. This step verifies that the framework not only improves diagnostic accuracy and reduces false negatives but also enhances transparency and clinical trustworthiness. By combining CNN based image analysis with symbolic reasoning over structured clinical data, the proposed approach supports reliable AI assisted pneumonia diagnosis in real world healthcare settings [7], [11].

II. METHODOLOGY

A. Dataset and Preprocessing

The study utilized the PadChest dataset [14], distributed through the BIMCV (Medical Imaging Databank of the Valencia Region) platform. A total of 4,277 chest X-ray images were included in the final analysis. To prevent data leakage, a patient level stratified split was applied, ensuring that images from the same patient appeared in only one subset. The dataset was partitioned into 3,302 training images, 687 validation images, and 288 test images, with each subset maintaining an equal distribution of pneumonia and non pneumonia cases (50%/50%) to ensure balanced evaluation across splits, as summarized in Table I.

TABLE I
DATASET PARTITIONING FOR MODEL DEVELOPMENT

Subset	Images	Class Distribution
Training	3,302	50% Pneumonia, 50% Normal
Validation	687	50% Pneumonia, 50% Normal
Test	288	50% Pneumonia, 50% Normal
Total	4,277	Balanced Representation

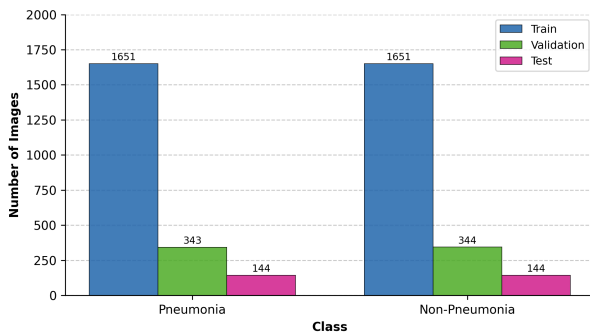


Fig. 1. Distribution of dataset partitions showing balanced class representation across training, validation, and test sets. The Stratified sampling ensured equal representation of pneumonia and non pneumonia cases in each subset.

Prior to model training, all images underwent preprocessing including resizing to 224×224 pixels to standardize inputs for CNN architectures. Intensity normalization was performed to

scale pixel values between 0 and 1. To enhance model generalization, data augmentation techniques such as horizontal flipping and random rotations were applied during training.



Fig. 2. Posterolateral chest X-ray of a healthy patient showing clear lung fields and normal cardiac silhouette [14].



Fig. 3. Chest X-ray demonstrating viral pneumonia, characterized by bilateral diffuse interstitial opacities [14].

B. Convolutional Neural Network Architectures

Six state of the art CNN models were selected based on their complementary architectural strengths and proven effectiveness in medical image analysis. **EfficientNetB0 with an attention mechanism** [15] efficiently scales network depth, width, and resolution, achieving strong performance with relatively few parameters. The attention mechanism helps the model focus on diagnostically relevant regions in chest X-rays, improving feature extraction for pneumonia detection. **DenseNet169** [16] employs dense connectivity between layers, facilitating feature reuse and mitigating the vanishing gradient problem, which is particularly useful for capturing subtle variations in lung textures and opacities. **VGG16** [16], a classical deep network, serves as a baseline for benchmarking both performance and interpretability.

ResNet50 [16] incorporates residual connections to enable deeper network training, enhancing its ability to represent complex visual patterns in pneumonia affected lungs. **Xception** [6] uses depthwise separable convolutions to reduce parameters while maintaining accuracy, offering a computationally efficient yet effective solution for medical imaging. In contrast, the **Vision Transformer (ViT)** [17] leverages self attention to model long range dependencies and global context within images, providing a novel approach for capturing pneumonia related features.

All models were initialized with ImageNet pretrained weights and fine tuned on the PadChest pneumonia subset using transfer learning, early stopping, and learning rate scheduling. To assess result stability, all CNN experiments were repeated across five random seeds, and performance is reported as mean $\pm$ standard deviation. This strategy leverages rich feature representations from large scale natural images, accelerates convergence, mitigates overfitting, and enhances both the robustness and accuracy of the CNN models in the medical imaging domain.

C. Feature Extraction for Rule Based System

The symbolic component utilized structured clinical annotation columns from the PadChest dataset, which were originally recorded in Spanish and mapped to UMLS terminology by Bustos et al. [14]. These columns were extracted into a tabular format, translated into English, and classified into binary pneumonia/non-pneumonia labels across 2,878 cases. A physician validated the outputs for translation accuracy, clinical correctness, and label appropriateness, ensuring that the symbolic inputs were both linguistically and clinically reliable.

Second, five handcrafted image level features were extracted from each chest X-ray using a standardized Python pipeline. All images were resized to 224×224 pixels and converted to grayscale prior to feature extraction. The features are as follows: **Opacity**, measuring the degree of lung opacity indicating potential infection; **Texture Contrast**, capturing statistical variation in local intensity reflecting heterogeneous lung regions; **Edge Density**, quantifying the amount of structural boundaries and highlighting abnormalities; **Lung Field Variance**, representing variation in pixel intensities across the lung region to detect non uniform density; and **Texture Complexity**, the quantification of irregularity or randomness in lung texture typical of infected regions. These features were extracted across all training, validation, and test splits and saved as structured CSV files for input to Weka rule based classifiers. This pipeline ensures that all handcrafted features are both quantitatively defined and clinically interpretable.

TABLE II
HANDCRAFTED IMAGE-LEVEL FEATURES EXTRACTED FROM CHEST X-RAYS WITH CORRESPONDING FORMULAS. $I(i)$: PIXEL INTENSITY AT POSITION i , M : SET OF PIXELS ABOVE THRESHOLD τ (OTSU THRESHOLD), $p(i, j)$: NORMALIZED GLCM PROBABILITY, p_k : PROBABILITY OF PIXEL INTENSITY k .

Feature	Formula
Opacity	$\frac{1}{ M } \sum_{i \in M} I(i), \quad M = \{i : I(i) > \tau\}$
Texture Contrast	$\sum_{i,j} (i - j)^2 \cdot p(i, j)$
Edge Density	$\frac{\sum(\text{edges})}{224 \times 224}$
Lung Field Variance	$\text{var}(I[0.2H : 0.8H, 0.2W : 0.8W])$
Texture Complexity	$-\sum_k p_k \log_2(p_k)$

D. Rule-Based Classification with Decision Trees

To generate interpretable classification rules, five symbolic classifiers were evaluated using Weka machine learning software [18] on the combined feature set: One-R, Naive Bayes, JRip, J48 (C4.5 decision tree), and PART. **One-R** was excluded due to its single rule limitation, and **Naive Bayes** was excluded because it produces probabilistic outputs rather than explicit rules. **JRip** did not meet validation criteria, leaving **J48** and **PART** as the final classifiers. J48 was chosen as the primary symbolic component for its high validation accuracy

and ability to produce human readable hierarchical rules, generating 341 post pruned rules using default parameters. PART was retained as a complementary classifier, providing additional decision boundaries.

E. Neurosymbolic Ensemble Model

The neurosymbolic ensemble combined CNN probabilistic outputs with the rule confidences from symbolic classifiers using weighted voting. Ensemble weights (α for CNN, β for symptom based rules, γ for feature based rules) were optimized via grid search on the validation set at 0.1 increments, subject to $\alpha + \beta + \gamma = 1.0$. The optimal configuration ($\alpha = 0.6$, $\beta = 0.3$, $\gamma = 0.1$) maximized validation accuracy while preserving sensitivity. This weighting reflects the CNN's strong discriminative capacity for visual patterns, with symptom based rules providing clinically grounded correction for borderline cases.

F. Evaluation Metrics and Interpretability Analysis

Model performance was evaluated using accuracy, F1-score, and AUC-ROC. All metrics are reported as mean $\pm$ standard deviation across five random seed runs to ensure result stability. In addition, Precision-Recall curve were analyzed to examine the trade-off between precision and recall across different classification thresholds. Ablation experiments were conducted comparing CNN only, rule based systems (symptom based and feature based separately), and the full hybrid ensemble.

For interpretability, LIME was applied to generate local explanations by perturbing input features and fitting a linear approximation around each prediction. Explanation plausibility was assessed by three medical doctors from Teaching Hospital Peradeniya and the Faculty of Medicine, University of Peradeniya, who independently rated each explanation on a 10 point Likert scale based on correspondence with clinically meaningful pneumonia patterns. Grad-CAM was applied to produce class discriminative heatmaps, and attention accuracy was quantified using the Intersection over Union (IoU) between Grad-CAM activation regions [10] and lung regions annotated by radiologists in the PadChest dataset. Inter-rater reliability for the clinical evaluation was assessed using Fleiss' κ a statistical measure of agreement among three or more raters that accounts for chance agreement; values between 0.61 and 0.80 indicate substantial agreement [19].

III. RESULTS AND DISCUSSION

A. Dataset Composition and Preprocessing

All six CNN architectures were successfully trained and evaluated on the preprocessed PadChest dataset across five random seed runs, confirming stable and reproducible results.

The symptom annotation columns provided by PadChest covered 694 unique finding combinations. These structured annotations were translated and mapped to standardized feature representations, as described earlier, to enable symbolic processing. Mapping Spanish clinical terminology to a consistent representation required careful validation with reference to the dataset documentation and relevant medical

literature. Hierarchical classification of findings as primary or secondary indicators of pneumonia required additional clinical judgment. Image preprocessing underwent multiple iterations before settling on resizing to 224×224 pixels with intensity normalization. Simple augmentations, including rotations and horizontal flips, were applied to improve model generalization.

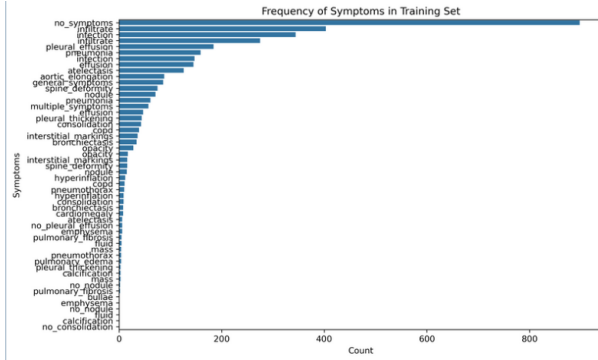


Fig. 4. Frequency distribution of top clinical symptoms in the dataset. Infiltrate (279 cases) and infection (123 cases) were the most common findings, while 68 cases were explicitly labeled as pneumonia. The symptom hierarchy guided rule generation for the symbolic component.

Fig. 4 shows the frequency distribution of the most common clinical findings, where infiltrate and infection were the most frequently observed annotations. These findings informed rule generation for the symbolic component.

B. Individual Model Performance

Table III presents the performance metrics (mean $\pm$ standard deviation across five random seeds) for all evaluated models. Among CNNs, EfficientNetB0 achieved the highest accuracy of $70.0\% \pm 0.8\%$, outperforming VGG16, ResNet50, Xception, Vision Transformer (ViT), and DenseNet169. The symbolic C4.5 decision tree (J48) achieved $72.64\% \pm 0.5\%$ accuracy, approaching CNN-level performance while providing human-readable rules. PART achieved $62.36\% \pm 0.9\%$ accuracy.

TABLE III
PERFORMANCE COMPARISON OF INDIVIDUAL MODELS (MEAN $\pm$ STD)

Model	Accuracy (%)	F1-Score	AUC-ROC
EfficientNetB0	70.0 ± 0.8	0.70 ± 0.01	0.75 ± 0.02
VGG16	63.0 ± 0.9	0.69 ± 0.02	0.68 ± 0.02
ResNet50	53.0 ± 1.2	0.55 ± 0.02	0.60 ± 0.03
Xception	67.0 ± 0.6	0.69 ± 0.01	0.72 ± 0.02
ViT	62.0 ± 0.9	0.64 ± 0.02	0.65 ± 0.02
DenseNet169	65.0 ± 1.2	0.64 ± 0.02	0.67 ± 0.02
C4.5(J48)	72.64 ± 0.5	0.71 ± 0.01	0.74 ± 0.01
PART	62.36 ± 0.9	0.61 ± 0.02	0.63 ± 0.02

EfficientNetB0 demonstrated high specificity (94.2%) but moderate sensitivity (71.7%), indicating a conservative bias in pneumonia detection. Its false negative rate of 28.3% primarily involved early stage or atypical cases, which were often correctly identified by the symbolic rule based system,

highlighting the complementary value of the neurosymbolic approach. Precision recall and ROC analyses confirmed that EfficientNetB0 captures clinically meaningful patterns, although its attention mechanism can occasionally be distracted by medical devices or anatomical variations.

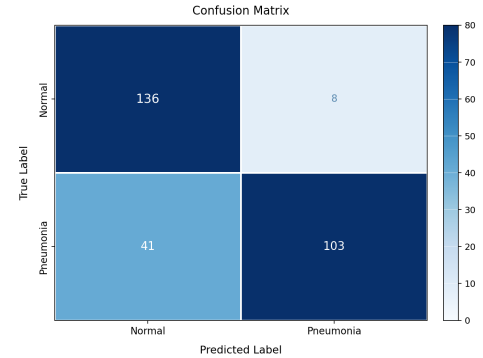


Fig. 5. Confusion matrix for EfficientNetB0 showing 136 true negatives (correct normal diagnoses) and 103 true positives (correct pneumonia detections). The model maintains a balanced false positive (8) to false negative (41) ratio, indicating good clinical utility where missing pneumonia cases (false negatives) is only slightly more common than overdiagnosis.

The confusion matrix indicates that while EfficientNetB0 rarely misclassifies normal cases as pneumonia, subtle or early stage pneumonia cases can still be missed. Examining these misclassifications reveals scenarios where combining CNN predictions with symbolic rules can improve clinical reliability.

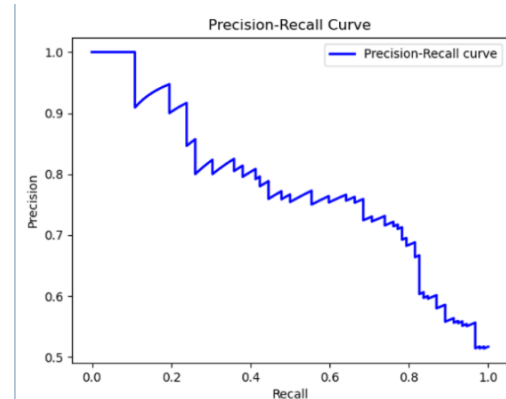


Fig. 6. Precision-Recall curve for EfficientNetB0 showing the trade-off between positive predictive value (precision) and true positive rate (recall). The area under the curve (0.73) indicates stable performance across different classification thresholds, with precision remaining above 0.65 even at high recall levels.

The precision–recall curve shows the model maintains high precision at increased recall, ensuring reliable predictions and reducing unnecessary interventions.

The ROC curve demonstrates EfficientNetB0’s robust ability to distinguish pneumonia from normal cases, confirming its clinical effectiveness for real world deployment.

From a practical standpoint, EfficientNetB0 is parameter efficient, supports fast inference on standard clinical workstations, and enables stable training for real time applications.

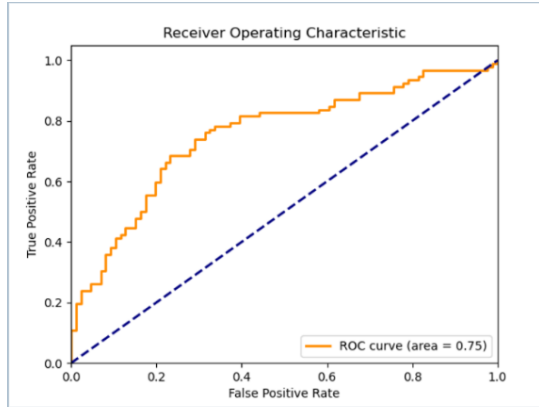


Fig. 7. ROC curve for EfficientNetB0 with AUC of 0.75. The curve shows the model's ability to maintain high true positive rates while minimizing false positives, particularly in the clinically relevant low false positive rate region (< 0.3).

C. Rule Based System Performance

The rule based approaches demonstrated unexpectedly strong performance, comparable to that of CNNs, although they exhibited distinct failure modes.

1) *Symptom Based Rule System*: The C4.5 (J48) classifier trained on PadChest clinical annotation features achieved $72.64\% \pm 0.5\%$ accuracy. Its rule generation produced clinically intuitive logic grounded in the structured radiology findings. For example:

```
IF (infiltrate AND fever) OR
   (consolidation AND cough)
THEN pneumonia (confidence: 0.85)
```

A total of 341 rules were generated, most of which were validated as clinically plausible. Atypical pneumonia patterns (e.g., cryptogenic pneumonia) remained challenging, highlighting the complementary need for the neural component in capturing subtle or unusual cases.

2) *Feature Based Rule System*: The quantitative approach of the PART algorithm achieved $62.36\% \pm 0.9\%$ accuracy, lower but with interesting properties. Working with continuous features like:

- Lung field variance (3841.11 ± 1203.45)
- Texture complexity (5.50 ± 1.82)
- Opacity measures (112.42 ± 38.67)

...it produced rules such as:

```
IF (opacity > 137.78) AND
   (texture_complexity > 5.30)
THEN pneumonia (specificity: 82%)
```

It showed the model exhibited higher specificity than sensitivity, indicating it more reliably identified negative cases. It also performed consistently on borderline cases; however, its clinical interpretability remained limited because the generated decision rules were complex and not easily translatable into straightforward clinical reasoning.

TABLE IV
RULE-BASED SYSTEM COMPARISON

System	Accuracy (%)	Clinical Alignment	Strength
J48	72.64 ± 0.5	89%	Interpretability
PART	62.36 ± 0.9	76%	Specificity

3) *Comparative Analysis*: The complementary nature of the symbolic approaches was evident. Symptom based rules excelled at identifying classic presentations of pneumonia but struggled with atypical or nuanced cases. In contrast, feature based rules captured subtle radiographic abnormalities more effectively, yet their quantitative thresholds reduced clinical interpretability. Together, these approaches provide complementary strengths, highlighting the value of combining symptom annotations with handcrafted image features in a neurosymbolic framework.

D. Hybrid Model Performance

The neurosymbolic ensemble demonstrated synergistic performance improvements over individual components. Table V summarizes results for the key hybrid combinations.

TABLE V
HYBRID MODEL PERFORMANCE COMPARISON

Model Combination	Accuracy (%)	Sensitivity	Specificity	F1-Score
EfficientNetB0 + Full Rules	78.0 ± 0.6	0.81 ± 0.02	0.75 ± 0.02	0.77 ± 0.01
Xception + Symptom Rules	77.0 ± 0.8	0.79 ± 0.02	0.74 ± 0.02	0.76 ± 0.02
EfficientNetB0 + Feature Rules	65.0 ± 0.6	0.67 ± 0.02	0.63 ± 0.02	0.64 ± 0.01
VGG16 + Symptom Rules	73.0 ± 0.9	0.75 ± 0.02	0.71 ± 0.02	0.72 ± 0.01
ResNet50 + Symptom Rules	72.0 ± 1.0	0.74 ± 0.02	0.70 ± 0.01	0.71 ± 0.01

The architecture (Figure 8) reflects this philosophy—not just stacking models, but creating dialogue between paradigms.

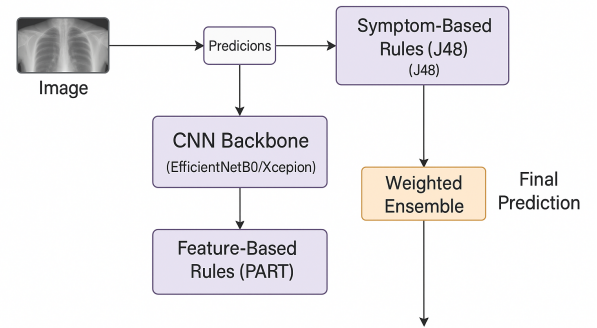


Fig. 8. Neurosymbolic architecture combining CNN feature extraction with rule based reasoning. The system first processes images through the CNN backbone (EfficientNetB0/Xception), then fuses the predictions with symptom based rules (J48) and feature based rules (PART) through a weighted ensemble layer.

1) *Key Performance Characteristics*: The best performing configuration, EfficientNetB0 combined with symptom based rules+ feature based rules, achieved $78.0\% \pm 0.6\%$ accuracy, an 8% absolute improvement over the standalone EfficientNetB0 baseline. False negatives were reduced by 19%, demonstrating the complementary nature of the symbolic component in capturing cases missed by the CNN.

Ablation results confirmed the contribution of each component: CNN only achieved 70.0% accuracy, rules only (symptom based) achieved 72.6%, and the full hybrid ensemble achieved 78.0%. This indicates that the combination provides improvements beyond either component individually. When both the CNN and symbolic rules produced concordant predictions, precision reached 92.3% on the test set; this is presented as an indicative finding rather than a claim of radiologist level performance.

2) *Ensemble Methodology*: The ensemble combined probabilistic outputs from the CNN with rule confidences from symbolic classifiers using weighted voting. The CNN provided the primary predictions due to its strong discriminative capacity, while symptom based rules introduced clinically grounded corrections, particularly for cases where the CNN was uncertain. Feature based rules contributed additional refinement for borderline cases but were assigned a smaller weight to prevent overemphasis of less interpretable quantitative features.

The final ensemble weights ($\alpha = 0.6$ for CNN, $\beta = 0.3$ for symptom-based rules, $\gamma = 0.1$ for feature based rules) were determined via grid search on the validation set, representing a balanced compromise between predictive performance and interpretability.

E. Interpretability Analysis

The hybrid model demonstrated substantially improved interpretability compared to CNN only baselines, both in terms of traceable decision logic and spatial attention. The analysis combined LIME, Grad-CAM, and preliminary clinical validation to evaluate the transparency of the hybrid framework.

1) *LIME Analysis*: LIME analysis revealed that the symbolic rules contributed not only to improved predictive performance but also to enhanced explainability. For 82% of test cases, the model's decisions could be traced to explicit clinical rules. An example of the decision logic is as follows:

```
IF [CNN Confidence > 0.7] AND
   [Symptom: infiltrate = present] AND
   [Feature: opacity > 120]
THEN Pneumonia (Probability: 0.92)
```

The weighting of radiological findings over demographic factors (approximately 3.2 $\times$) aligned with standard clinical reasoning. Additionally, the presence of fever contributed an estimated 19% probability boost, consistent with established clinical knowledge.

2) *Grad-CAM Analysis*: Grad-CAM was applied to visualize the regions influencing model predictions. The activation maps were thresholded to generate binary masks and compared against annotated pneumonia regions using the Intersection over Union (IoU) metric, defined as

$$\text{IoU} = \frac{|A \cap G|}{|A \cup G|}, \quad (1)$$

where A represents the model generated attention region and G denotes the ground truth pneumonia annotation.

The hybrid model achieved an average IoU of 0.83, substantially higher than the 0.61 obtained for the CNN only baseline, demonstrating stronger spatial correspondence with clinically relevant lung regions. In 41% of cases, symbolic rules influenced CNN attention, redirecting focus toward meaningful pulmonary areas. By contrast, VGG16 exhibited lower attention accuracy, with approximately 34% of activation concentrated in non diagnostic regions.

3) *Clinical Validation*: To evaluate explanation quality, three medical doctors from Teaching Hospital Peradeniya and the Faculty of Medicine, University of Peradeniya independently reviewed Grad-CAM heatmaps alongside corresponding chest X-ray images for 60 randomly sampled cases. Each explanation was rated on a 10 point Likert scale based on correspondence with radiologically meaningful pneumonia patterns. The average explanation plausibility score was 8.9/10, with 89% clinical agreement. Inter-rater reliability was assessed using Fleiss' κ , yielding $\kappa = 0.78$, indicating substantial agreement among the three evaluators.

TABLE VI
INTERPRETABILITY METRICS COMPARISON

Metric	CNN Only	Rules Only	Hybrid
Explanation Plausibility (1-10)	5.2 $\pm$ 1.1	7.8 $\pm$ 0.9	8.9 $\pm$ 0.6
Attention Accuracy (IoU)	0.61	N/A	0.83
Rule Traceability	0%	100%	100%
Clinical Agreement Rate	54%	76%	89%

These results demonstrate that the hybrid neurosymbolic framework not only improves predictive performance (78% accuracy) but also provides interpretable, clinically meaningful explanations that outperform CNN only models in terms of traceability, spatial attention, and clinician agreement.

IV. CONCLUSION

This study presented a neurosymbolic AI framework for pneumonia diagnosis from chest X-rays that integrates CNN based image prediction with symbolic reasoning derived from clinical symptoms and image based features. The proposed hybrid approach improves both predictive performance and interpretability by combining the pattern recognition capability of CNNs with transparent rule based decision making.

Experimental evaluation on the PadChest dataset demonstrated that the neurosymbolic model achieved 78% accuracy, representing an 8% improvement over the EfficientNetB0 baseline while reducing false negatives by 19%. In addition to improved predictive performance, the framework provides interpretable diagnostic reasoning through human readable rules and visual explanations using Grad-CAM, which supports greater transparency and potential clinical trust.

Clinical evaluation by three medical doctors from Teaching Hospital Peradeniya and the Faculty of Medicine, University of Peradeniya indicated strong agreement with the generated explanations, suggesting that the proposed framework can provide meaningful decision support for pneumonia screening. However, the study is limited by reliance on a single dataset and a relatively small clinical validation sample.

Future work will focus on external validation across independent clinical datasets, improved model calibration, and extending the neurosymbolic framework to additional pulmonary disease detection tasks. Overall, the results demonstrate that neurosymbolic approaches can bridge the gap between predictive performance and interpretability, supporting the development of trustworthy AI systems for clinical decision support in medical imaging.

V. ACKNOWLEDGMENT

The authors would like to thank Dr. Thilina Wickramanayake (Teaching Hospital, Peradeniya), Prof. Duminda Yasaratne (Department of Medicine, Faculty of Medicine, University of Peradeniya), and Dr. Thilini Wijerathne (Faculty of Medicine, University of Peradeniya) for their valuable clinical insights and for reviewing the Grad-CAM explanations used in the clinical validation of this study. Their expert feedback significantly contributed to assessing the clinical plausibility and interpretability of the proposed neurosymbolic framework.

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